

0.0.1 Interacting Multipoles of Rank 1 and 3

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \end{array} \right] \quad (1)$$

Simplify deltas:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\delta}^A \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\gamma\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\beta\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\beta\beta\delta}^A \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\delta}^A \\ &-15 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\gamma\delta}^A \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \end{aligned} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\beta\gamma\delta}^A \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^B \cdot \Omega_{\beta\gamma\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\beta\beta\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\beta\beta\delta}^A \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\gamma}^A \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\delta}^A \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\alpha\gamma\delta}^A \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \\ &+3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \end{aligned} \right] \quad (2b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\beta^B \cdot \Omega_{\beta\gamma\gamma}^A \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\beta\beta\gamma}^A \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\beta\beta\gamma}^A \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\gamma}^A \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\gamma}^A \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\beta \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\gamma}^A \\ +3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \\ +3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \\ +3 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \end{array} \right] \quad (2c)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Omega_{\beta\gamma\delta}^A \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha^B \cdot \Omega_{\beta\gamma\gamma}^A \\ -30 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\beta\beta\gamma}^A \\ -45 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\gamma}^A \\ +9 \cdot R_z^4 \cdot \mu_\alpha^B \cdot \Omega_{\alpha\beta\beta}^A \end{bmatrix} \quad (2d)$$

Simplify R:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{zzzz}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{z\gamma\gamma}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Omega_{z\beta\beta}^A \\ -45 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_{\alpha}^B \cdot \Omega_{zz\alpha}^A \\ +9 \cdot R_z^4 \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_{\alpha}^B \cdot \Omega_{zz\alpha}^A \\ +9 \cdot R_z^4 \cdot \mu_{\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \end{bmatrix} \quad (3b)$$

Multiply Out Sum:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_x^B \cdot \Omega_{xzz}^A \\ -45 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ -45 \cdot R_z^4 \cdot \mu_z^B \cdot \Omega_{zzz}^A \\ +9 \cdot R_z^4 \cdot \mu_x^B \cdot \Omega_{xxx}^A \\ +9 \cdot R_z^4 \cdot \mu_x^B \cdot \Omega_{xyy}^A \\ +9 \cdot R_z^4 \cdot \mu_x^B \cdot \Omega_{xzz}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yyy}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ +9 \cdot R_z^4 \cdot \mu_z^B \cdot \Omega_{xxz}^A \\ +9 \cdot R_z^4 \cdot \mu_z^B \cdot \Omega_{yyz}^A \\ +9 \cdot R_z^4 \cdot \mu_z^B \cdot \Omega_{zzz}^A \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yyy}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \end{bmatrix} \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -36 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ +9 \cdot R_z^4 \cdot \mu_y^B \cdot \Omega_{yyy}^A \end{bmatrix} \quad (4c)$$

Combining everything:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = \begin{bmatrix} +\frac{12}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yzz}^A \\ -\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ -\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yyy}^A \end{bmatrix} \quad (5)$$

A:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \mu_{\alpha}^B \Omega_{\beta\gamma\delta}^A = \begin{bmatrix} -\frac{12}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yyy}^A \\ -\frac{12}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ -\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{xxy}^A \\ -\frac{3}{5} \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yyy}^A \end{bmatrix} \quad (6a)$$

$$\xrightarrow{\text{added up}} \begin{bmatrix} -3 \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{yyy}^A \\ -3 \cdot R_z^{-5} \cdot \mu_y^B \cdot \Omega_{xxy}^A \end{bmatrix} \quad (6b)$$